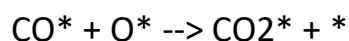
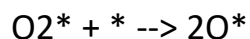
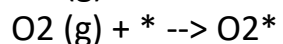
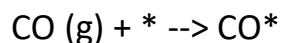


31 Linearly independent reactions

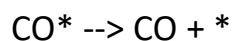
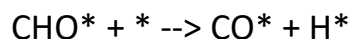
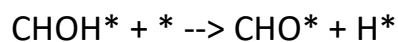
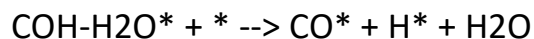
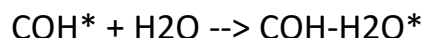
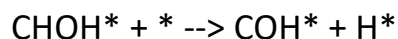
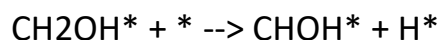
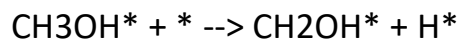
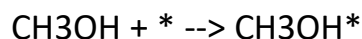
I found [this video](#) helpful when writing this lecture.

The following is an example of a linearly independent set of reactions :

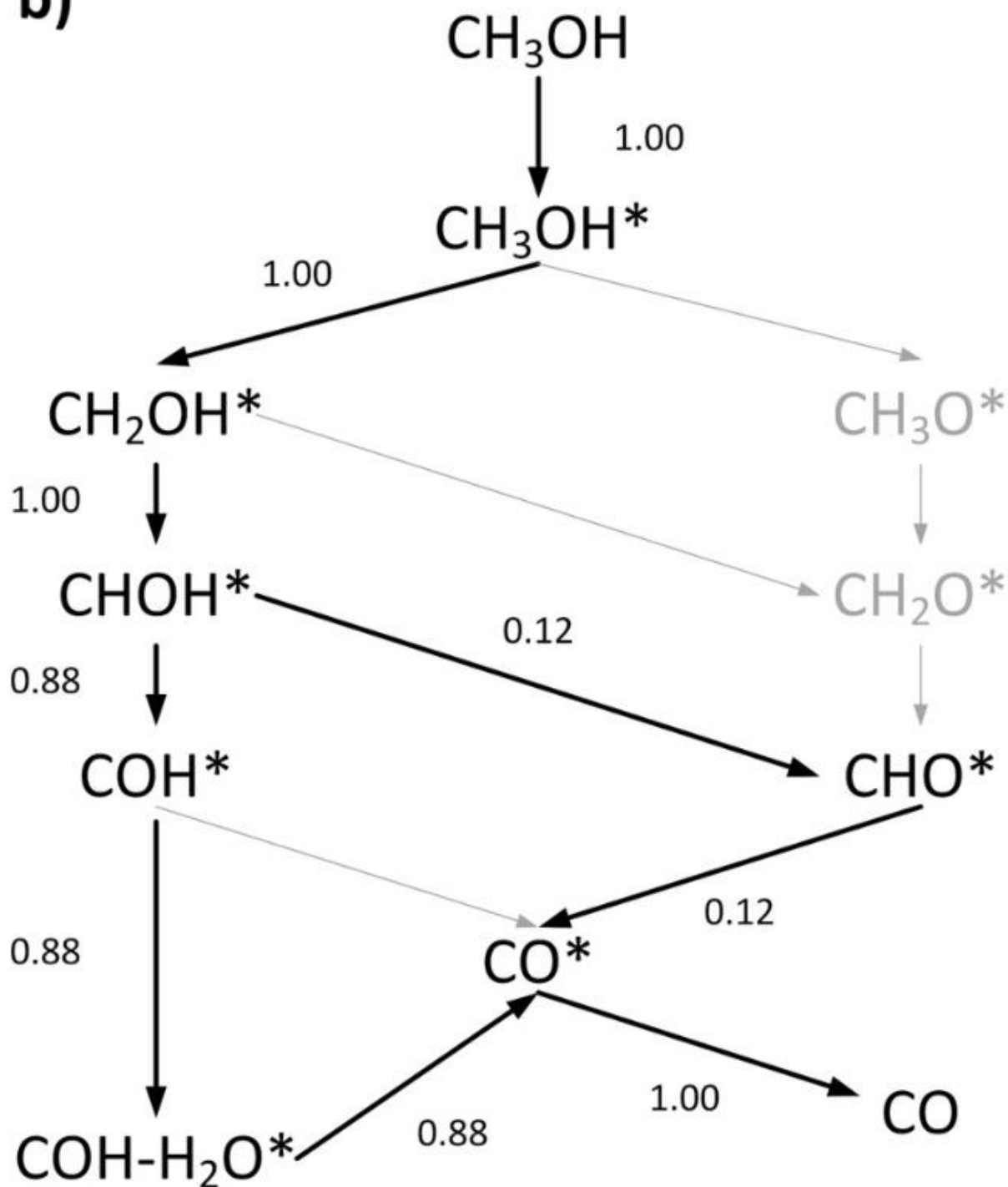


This mechanism represents a "straight path" from reactants to products, i.e., that there is only one route from reactants to products.

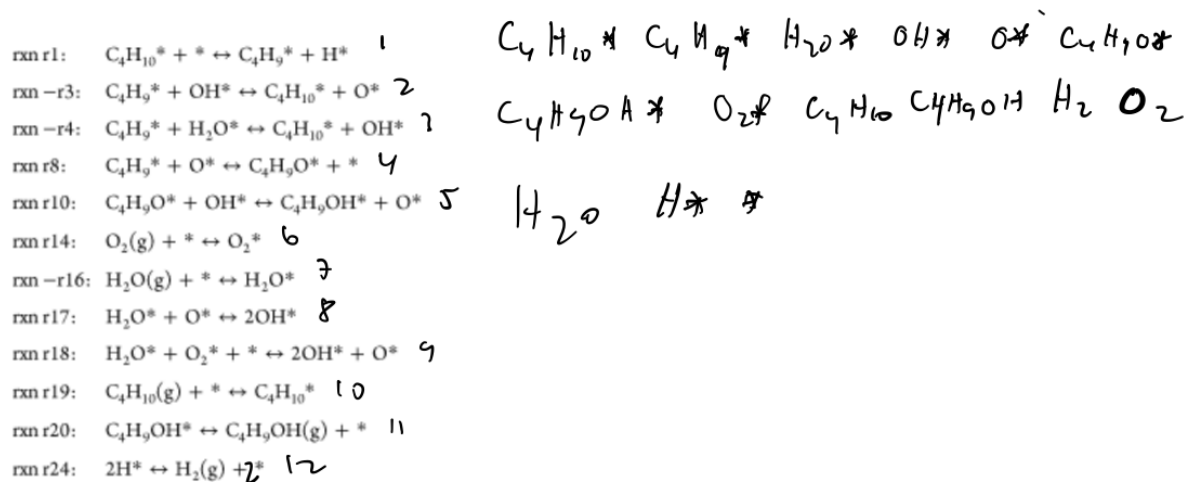
However even branched reaction mechanisms can include only linearly independent reactions. For example often involve branched paths, for example, [García Cárcamo et al](#) found the following mechanism (and corresponding reaction path) for methanol dehydrogenation over Pt/Al₂O₃ catalysts:



b)



In contrast, the mechanism of n-butane oxidation to 1-butanol by [Zhu et al](#), i.e.:



involves linearly dependent steps. It is totally **OK** to have a

mechanism where not all steps are linearly independent. In fact, this is how chemistry often operates. However, challenges arise when

- trying to identify the overall gas phase reaction, calculating the total reaction energy or free energy, selecting steps from which to write coverage or concentration relations, etc

Hence, we need to be able to know how many reactions within a mechanism are linearly independent. We can do this using the procedure of

Gaussian elimination

Let's do an example. This is Exercise 2.5 from Rawlings and Ekerdt: Consider a mixture of CO, H₂, and CH₄ that is fed into a furnace with O₂ and produces CO, CO₂, and H₂O. The following chemical reactions have been suggested to account for the products that form:

1. $\text{CO} + 1/2 \text{O}_2 \rightarrow \text{CO}_2$
2. $\text{H}_2 + 1/2 \text{O}_2 \rightarrow \text{H}_2\text{O}$
3. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
4. $\text{CH}_4 + 3/2 \text{O}_2 \rightarrow \text{CO} + 2\text{H}_2\text{O}$

How many linearly independent reactions are there? Which reaction cannot be eliminated from the set? Why?

First step: Write mechanism in matrix form

$$\begin{array}{c} \text{CO} \quad \text{O}_2 \quad \text{CO}_2 \quad \text{H}_2 \quad \text{H}_2\text{O} \quad \text{CH}_4 \\ \begin{array}{l} 1 \\ 2 \\ 3 \\ 4 \end{array} \left[\begin{array}{cccccc} -1 & -1/2 & 1 & 0 & 0 & 0 \\ 0 & -1/2 & 0 & -1 & 1 & 0 \\ 0 & -2 & 1 & 0 & 2 & -1 \\ 1 & -3/2 & 0 & 0 & 2 & -1 \end{array} \right] \end{array}$$

x 2 by 2

$$\begin{array}{c}
 1 \\
 2 \\
 3 \\
 4
 \end{array}
 \left[\begin{array}{cccccc}
 \text{CO} & \text{O}_2 & \text{CO}_2 & \text{H}_2 & \text{H}_2\text{O} & \text{CH}_4 \\
 -1 & -\frac{1}{2} & 1 & 0 & 0 & 0 \\
 0 & -1 & 0 & -2 & 2 & 0 \\
 0 & -2 & 1 & 0 & 2 & -1 \\
 1 & -\frac{3}{2} & 0 & 0 & 2 & -1
 \end{array} \right]$$

$$-2 \times 2 + 3$$

$$\begin{array}{c}
 1 \\
 2 \\
 3 \\
 4
 \end{array}
 \left[\begin{array}{cccccc}
 \text{CO} & \text{O}_2 & \text{CO}_2 & \text{H}_2 & \text{H}_2\text{O} & \text{CH}_4 \\
 -1 & -\frac{1}{2} & 1 & 0 & 0 & 0 \\
 0 & -1 & 0 & -2 & 2 & 0 \\
 0 & 0 & 1 & 4 & -2 & -1 \\
 1 & -\frac{3}{2} & 0 & 0 & 2 & -1
 \end{array} \right]$$

add $1 \frac{1}{2} \times 4$

$$\begin{array}{c}
 1 \\
 2 \\
 3 \\
 4
 \end{array}
 \left[\begin{array}{cccccc}
 \text{CO} & \text{O}_2 & \text{CO}_2 & \text{H}_2 & \text{H}_2\text{O} & \text{CH}_4 \\
 -1 & -\frac{1}{2} & 1 & 0 & 0 & 0 \\
 0 & -1 & 0 & -2 & 2 & 0 \\
 0 & 0 & 1 & 4 & -2 & -1 \\
 0 & -2 & 1 & 0 & 2 & -1
 \end{array} \right]$$

$-2 \times 2 + 4$

$$\begin{array}{c}
 \text{CO} \quad \text{O}_2 \quad \text{CO}_2 \quad \text{H}_2 \quad \text{H}_2\text{O} \quad \text{CH}_4 \\
 \begin{array}{l} 1 \\ 2 \\ 3 \\ 4 \end{array} \left[\begin{array}{cccccc} -1 & -\frac{1}{2} & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -2 & 2 & 0 \\ 0 & 0 & 1 & 4 & -2 & -1 \\ 0 & 0 & 1 & 4 & -2 & -1 \end{array} \right]
 \end{array}$$

subtract 3 from 4

$$\begin{array}{c}
 \text{CO} \quad \text{O}_2 \quad \text{CO}_2 \quad \text{H}_2 \quad \text{H}_2\text{O} \quad \text{CH}_4 \\
 \begin{array}{l} 1 \\ 2 \\ 3 \\ 4 \end{array} \left[\begin{array}{cccccc} -1 & -\frac{1}{2} & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -2 & 2 & 0 \\ 0 & 0 & 1 & 4 & -2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

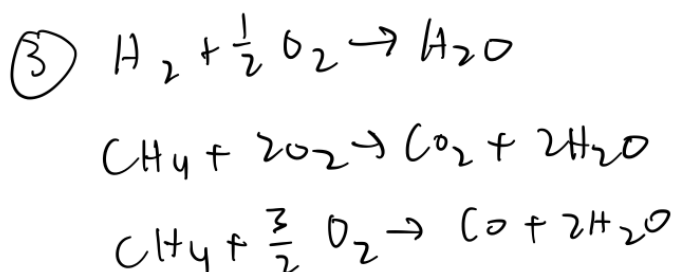
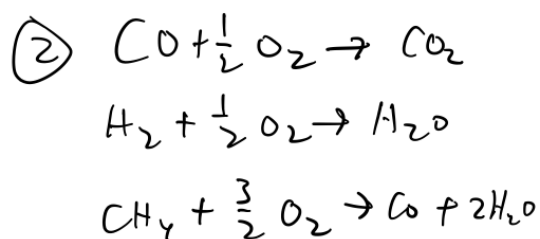
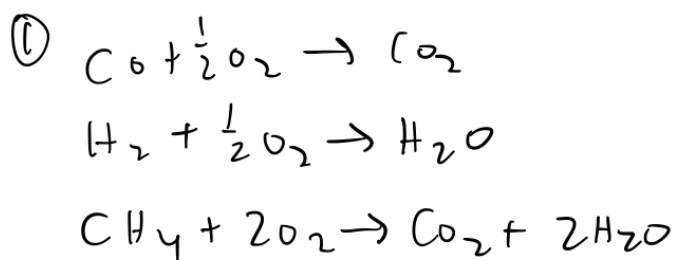
$\times 3$ by -1

$$\begin{array}{c}
 \text{CO} \quad \text{O}_2 \quad \text{CO}_2 \quad \text{H}_2 \quad \text{H}_2\text{O} \quad \text{CH}_4 \\
 \begin{array}{l} 1 \\ 2 \\ 3 \\ 4 \end{array} \left[\begin{array}{cccccc} -1 & -\frac{1}{2} & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -2 & 2 & 0 \\ 0 & 0 & -1 & -4 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

three independent rxns

rxn 2 cannot be eliminated, it's the only one that involves H_2

Possible sets are



What if you have a larger mechanism such as the one by Zhu et al? Use mathematical software to determine the matrix rank. For example, this is how it can be done in [python](#).